CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 7: Indices (Exponents)

Every formula and result of the chapter on one sheet

1	Product law	$a^m \times a^n = a^{m+n}$
2	Quotient law	$a^m \div a^n = a^{m-n}$
3	Power of a power	$(a^m)^n = a^{mn}$
4	Power of a product	$(ab)^n = a^n b^n$
5	Power of a quotient	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
6	Zero index	$a^0 = 1, a \neq 0$
7	Negative index	$a^{-n} = \frac{1}{a^n}$
8	Negative index of a fraction	$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$
9	Root	$a^{\frac{1}{n}} = \sqrt[n]{a}$
10	Fractional index	$a^{\frac{m}{n}} = \sqrt[n]{a^m}$
11	Equal powers, same base	$a^m = a^n$ gives $m = n$
12	Root of a root	$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$